Predicting Falcon 9 First Stage Landing Success

A Data-Driven Analysis for Launch Cost Estimation



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By Rhyz Manalo

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GitHub Project Repository

- https://github.com/rkdylan-manalo/IBM-Data-Science-Capstone-Project

OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
  - Visualisation – Charts
  - Folium Maps & Markers
  - Dashboard
  - Predictive Analysis
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EXECUTIVE SUMMARY



- Project Objective: Use Machine Learning model that predicts the successful landing of a Falcon 9 (first stage), enabling more accurate launch cost estimation.
- Methodology:
  1. Data Wrangling: Collected & cleaned historical launch data using SpaceX APIs & Web Scraping.
  2. Exploratory Data Analysis: Analysed factors influencing landing success, including launch site, payload mass, and orbit type.
  3. Interactive Visual Analytics: Created dashboards + maps to identify patterns & optimise launch site selection.
  4. Predictive Modelling: Trained & tuned four classification models to find the most accurate predictor (Logistic Regression, SVM, Decision Trees, KNN).
- Key Findings:
  - Falcon 9's landing success significantly increased over time.
  - Launch Sites: Success rates vary depending on location, with KSC LC-39A showing the best performance.
  - Best Model: The Decision Tree Model achieved 83.33% Test Accuracy (86.07% Cross-Validation).
- Recommendations:
  - Use the Decision Tree model to pre-launch predictions
  - This model can provide reliable cost estimates for potential clients, improving financial planning.

INTRODUCTION

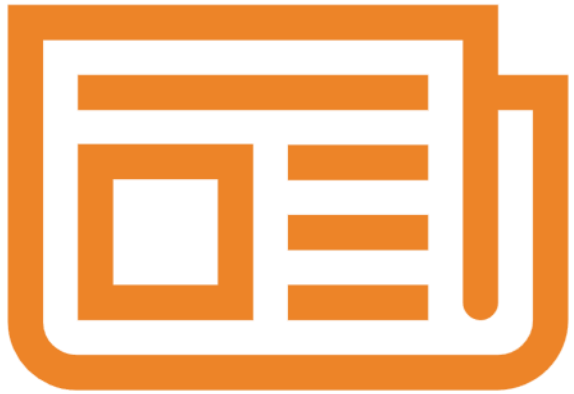


- Problem: SpaceX launch cost depends on whether the Falcon 9 first stage lands successfully and can be reused.
- Goal: Predict first-stage landing outcomes
- Key Questions:
  - Which factors influence landing success?
  - Which launch sites and orbits perform best?
  - Can machine learning accurately predict landing?
- Scope: Falcon 9 only (historical launch data)

METHODOLOGY: Data Collection & More

https://github.com/rkdylan-manalo/IBM-Data-Science-Capstone-Project/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

https://github.com/rkdylan-manalo/IBM-Data-Science-Capstone-Project/blob/main/jupyter-labs-webscraping.ipynb



- Data Sources:
  - SpaceX API: https://api.spacexdata.com/v4/launches/past/
  - Wikipedia Falcon 9 Launch History Tables
- Data Wrangling:
  - Parsed JSON/HTML => Pandas Dataframe
  - Sampled out Falcon 1 Data
  - Filled Payload Mass Null Values with the Mean
  - Launch Pad Null Values Kept (Represents No Lauch Pad Usage)
- Feature Engineering:
  - Mapped 8 Outcomes => Binary Y (0: Bad, 1: Good)
  - Used launch pad numbers, site, payload mass, booster version, etc.
- Analysis & Modelling:
  - EDA, Folium Maps, Plotly Dash, Dash Dashboard
  - Models: Logistic Regression, SVM, Decision Trees, KNN
  - GridSearch + Train Test Split + Confusion Matrix

Methodology: SQL Queries

https://github.com/rkdylan-manalo/IBM-Data-Science-Capstone-Project/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

- Display the names of the unique launch sites in the space mission: %sql SELECT DISTINCT Launch_Site FROM SPACEXTBL;
- Display 5 records where launch sites begin with the string 'CCA': %sql SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%'LIMIT 5;
- Display the total payload mass carried by boosters launched by NASA (CRS): %sql SELECT SUM(PAYLOAD_MASS__KG_) ASTOTAL_PAYLOAD_MASS FROM SPACEXTBL WHERE CUSTOMER = 'NASA (CRS)';
- Display average payload mass carried by booster version F9 v1.1: %sql SELECT AVG(PAYLOAD_MASS__KG_) ASAVERAGE_PAYLOAD_MASS FROM SPACEXTBL WHERE BOOSTER_VERSION = 'F9 v1.1';
- List the date when the first successful landing outcome in ground pad was achieved: %sql SELECT MIN(Date) ASFIRST_SUCCESSFUL_GROUND_LANDING FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000: %%sqlSELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_> 4000 AND PAYLOAD_MASS__KG_ < 6000;
- List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function: importsqlite3 import pandas as pd with sqlite3.connect("my_data1.db") as conn: result = pd.read_sql_query(""" SELECT DISTINCTBooster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL); """,conn) display(result)
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for themonths in year 2015: %%sql SELECT SUBSTR(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROMSPACEXTBL WHERE Landing_Outcome = 'Failure (drone ship)' AND SUBSTR(Date, 1, 4) = '2015';
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order: %%sql SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTBL WHERE DateBETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;

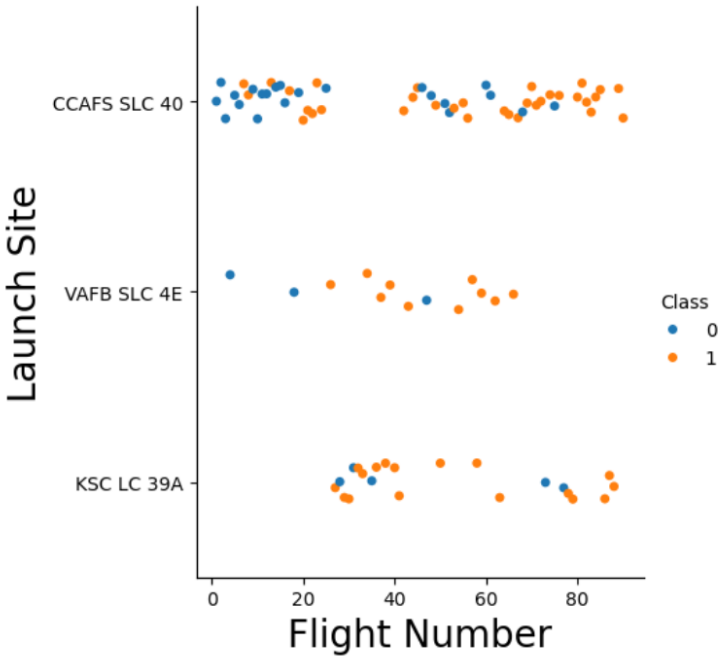
Methodology: Folium Maps & Markers

- Markers were put on each SpaceX launch site in North America, showing the total number of launches at each launch site
  - These markers help answer questions like "Are all launch sites near the coast?" and "Are all launch sites near the geographic equator?"
- Included labels of each SpaceX launch site on top/under each marker on the Folium Map
- PolyLine included to measure the distance of a launch site from the nearest coast.

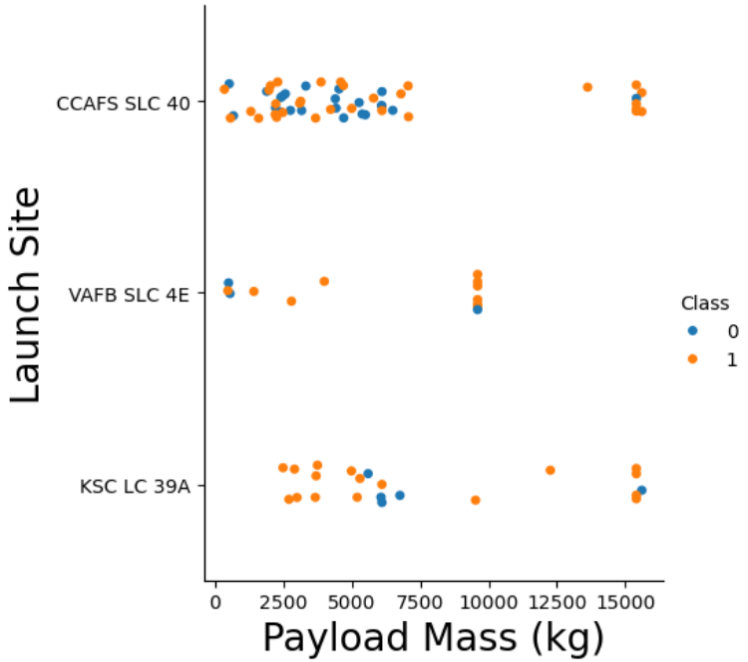
- https://github.com/rkdylan-manalo/IBM-Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

RESULTS: Launch Site Trends in Flight Numbers & Payload Mass (kg)

Takeaway: Success Rate Significantly Increases with Flight Number



Scatter Plot showing Flight Number vs Launch Site (Class)



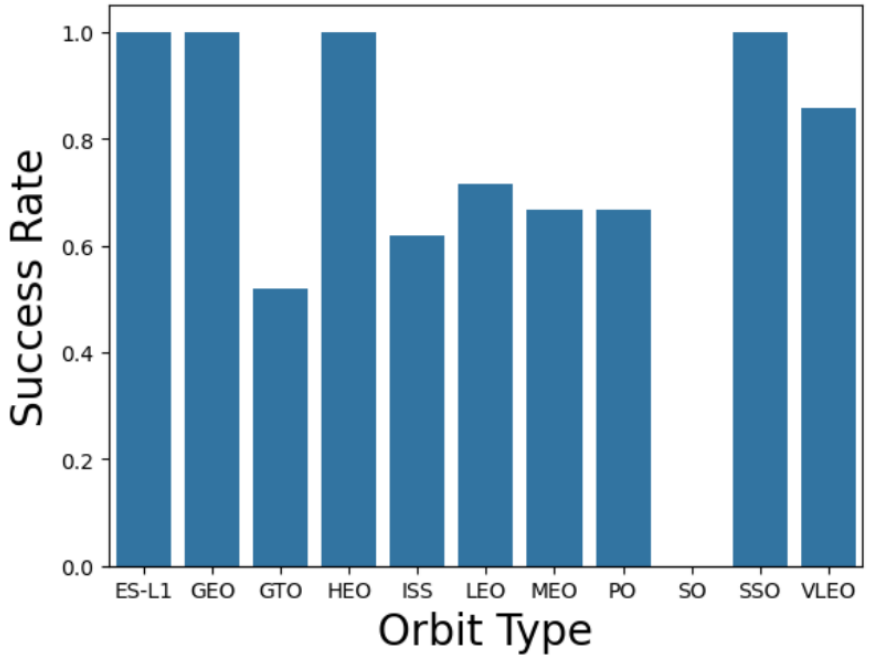
Scatter Plot showing Payload Mass (kg) vs Launch Site (Class)

Takeaway: Success Rate Varies with Payload Mass (kg) for Different Launch Sites

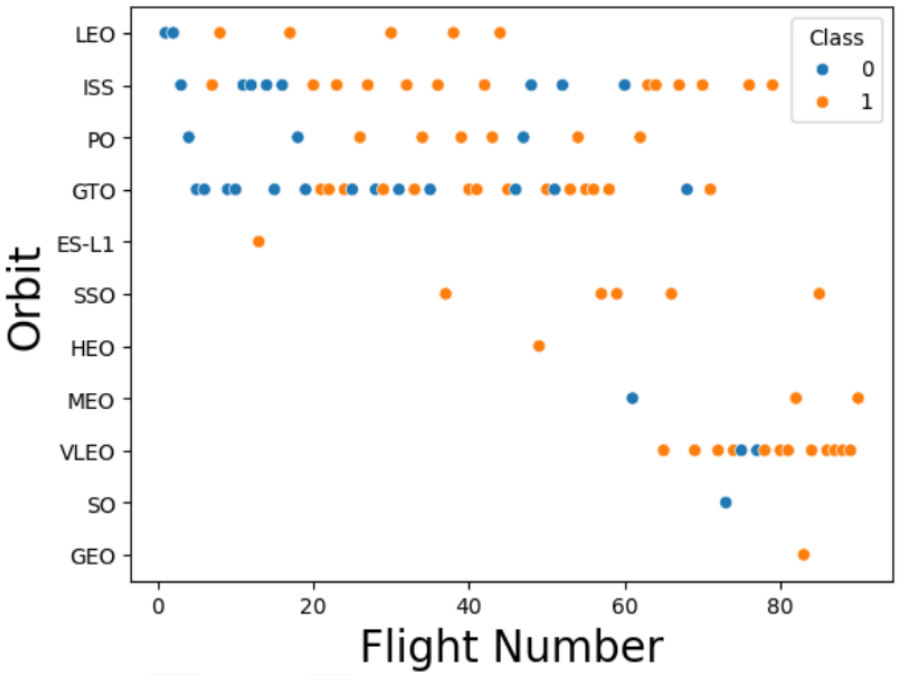
RESULTS: Orbit Type Success Rates & Flight History

Takeaway: ES-L1, GEO, HEO & SSO had a 100% success rate.

SO had the lowest success rate of 0%



Bar Chart Showing Success Rate of Different Orbit Types



Scatter Plot Showing Success Rate against Flight Number

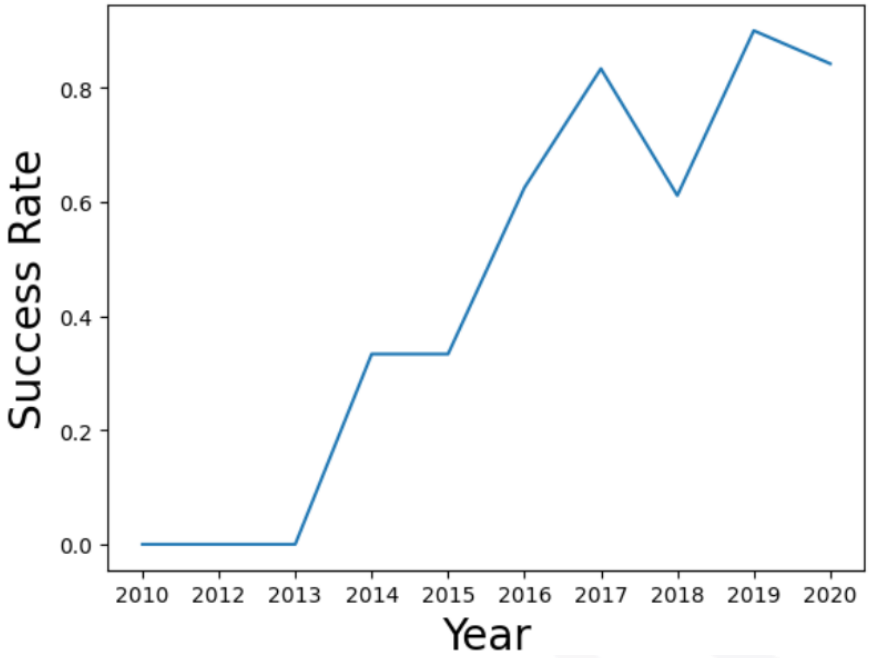
Takeaway: Flight Number Incraess Success Rate Regardless of Orbit Type (in most cases)

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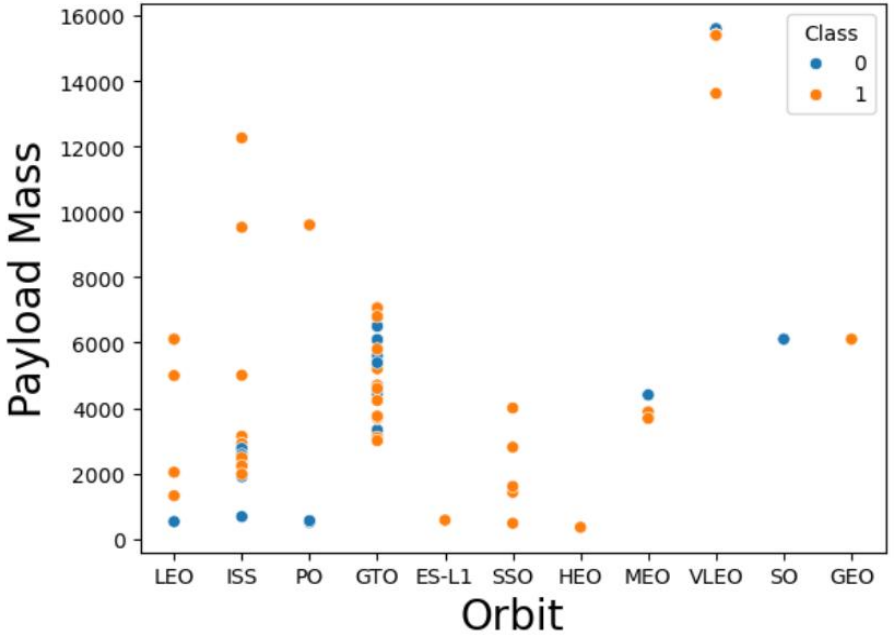
RESULTS: Overall Success Trends & Payload Mass by Orbit

Takeaway: Success Rate Significantly Incraesed from 2010-2020.

Sharpest Increases were in 2013, 2015 & 2016



Line Chart showing Success Rate of Launches from 2010-2020

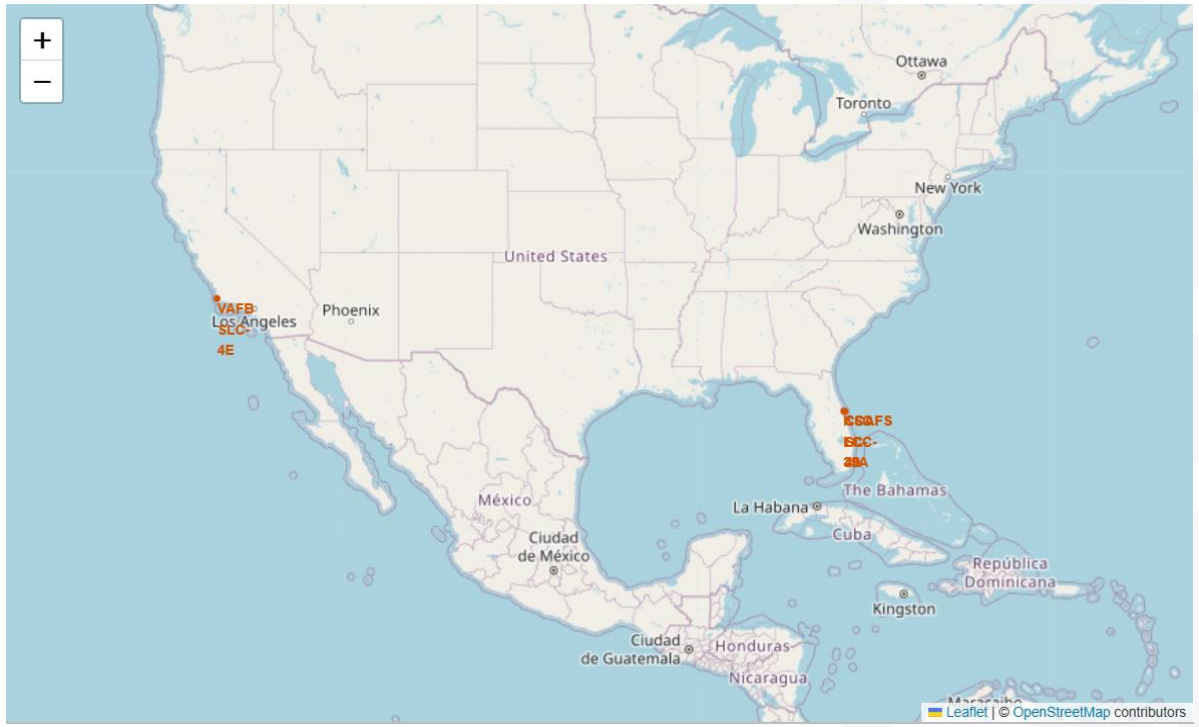


Scatter Plot showing Successes & Failures of Orbit Types versus Payload Mass (kg)

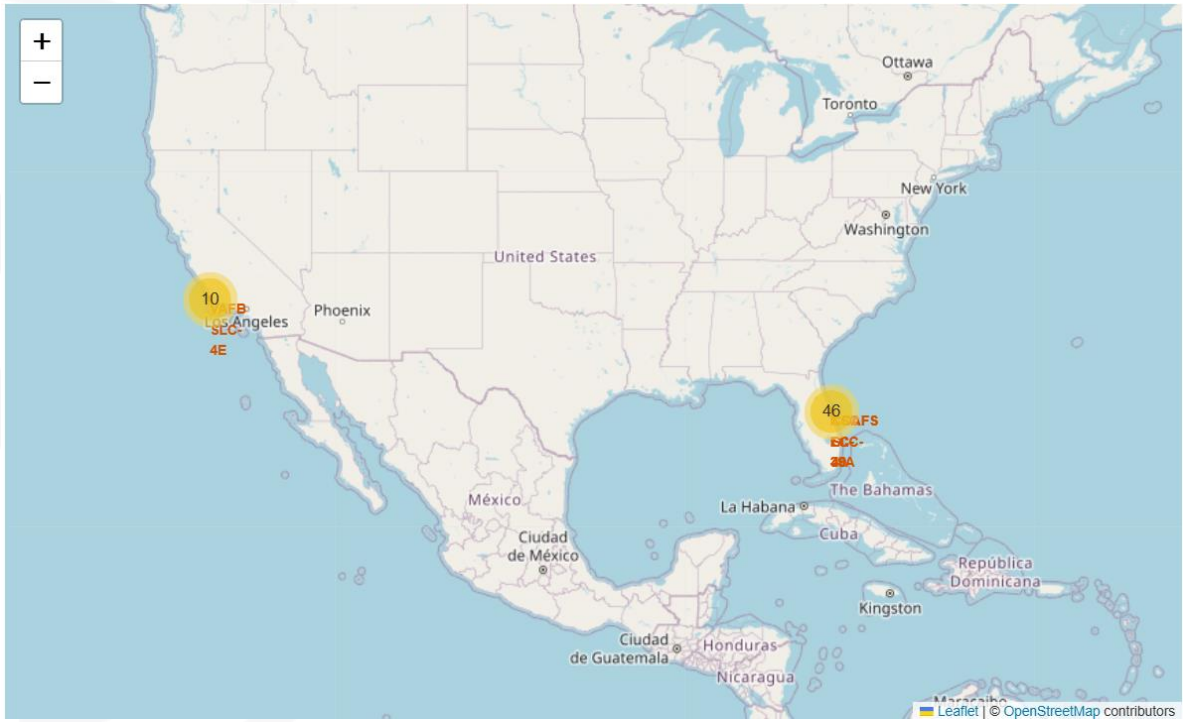
Takeaway: Success of Launches for Different Orbit Types varies with Payload Mass (kg)

RESULTS: Maps with SpaceX Launch Sites in North America

Takeaway: Optimal Launch Sites are Close to the Coast (for safety), and/or the Equator (Orbital Mechanics)

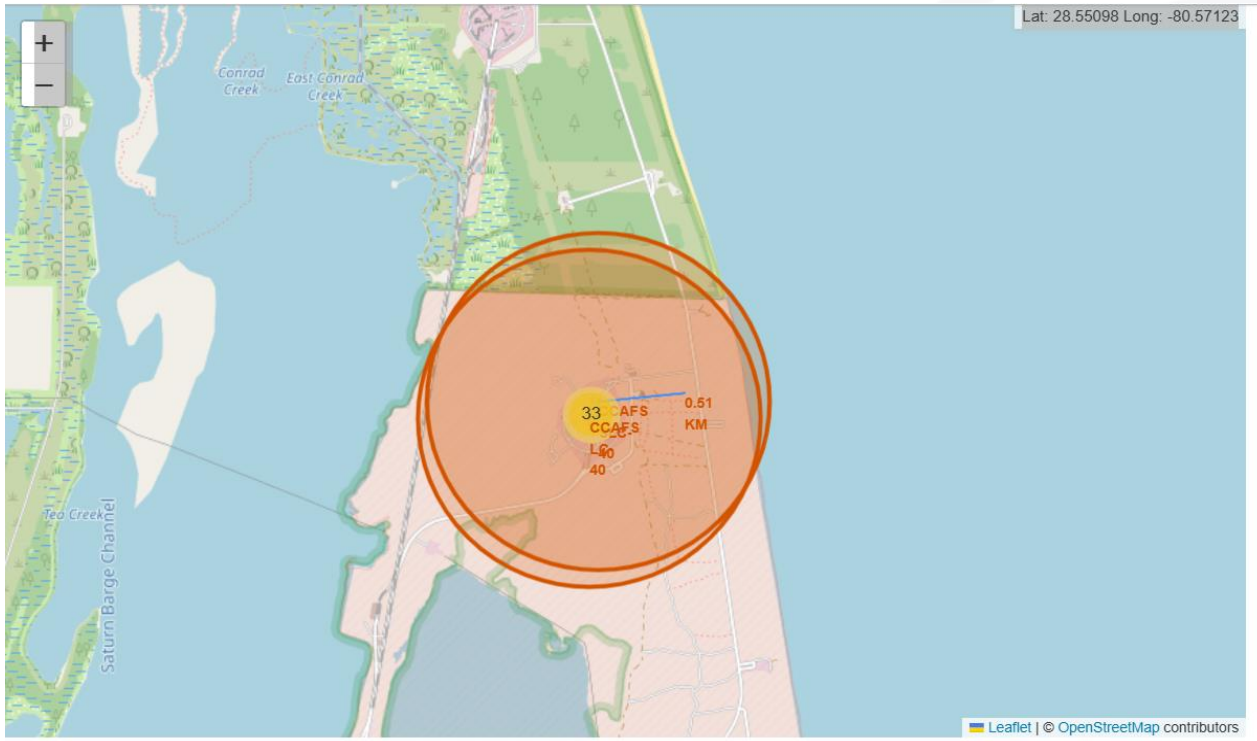


Map with Markers of SpaceX Launch Sites

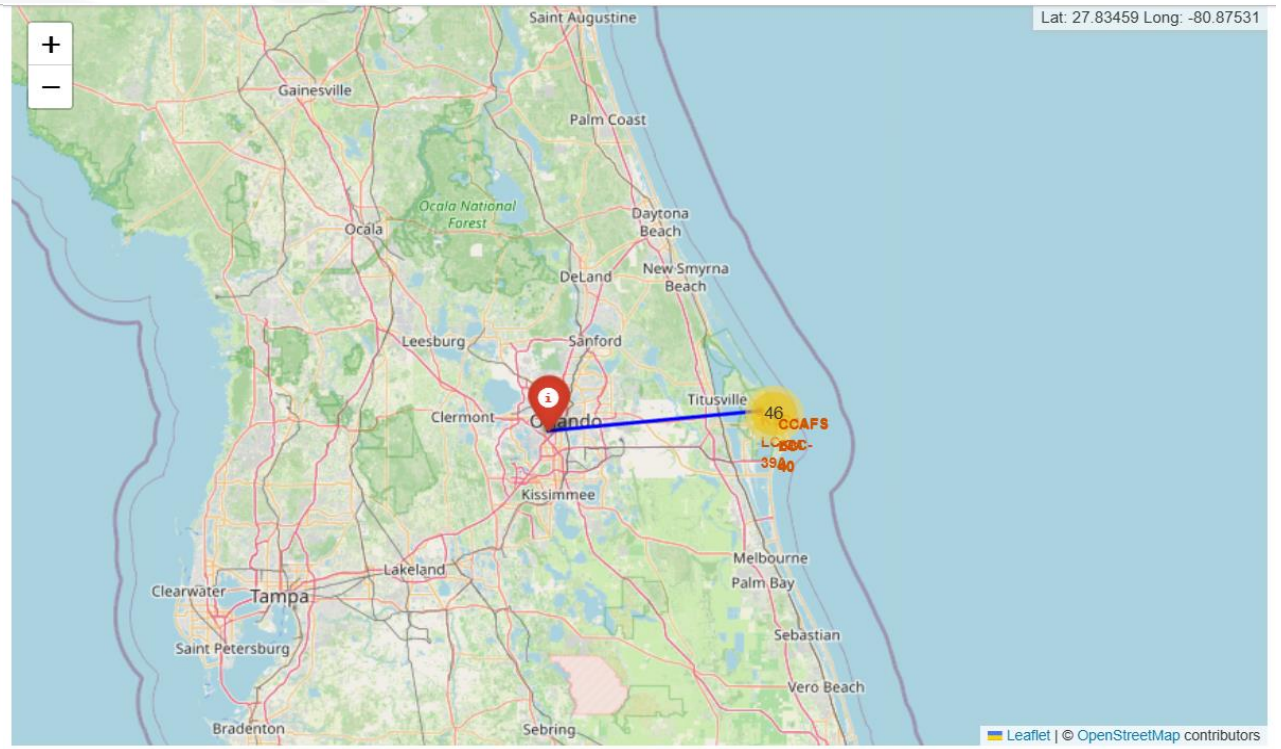


Map with Markers of SpaceX Launch Sites and Number of Launches at each Launch Site

DATABASE TRENDS - FINDINGS & IMPLICATIONS



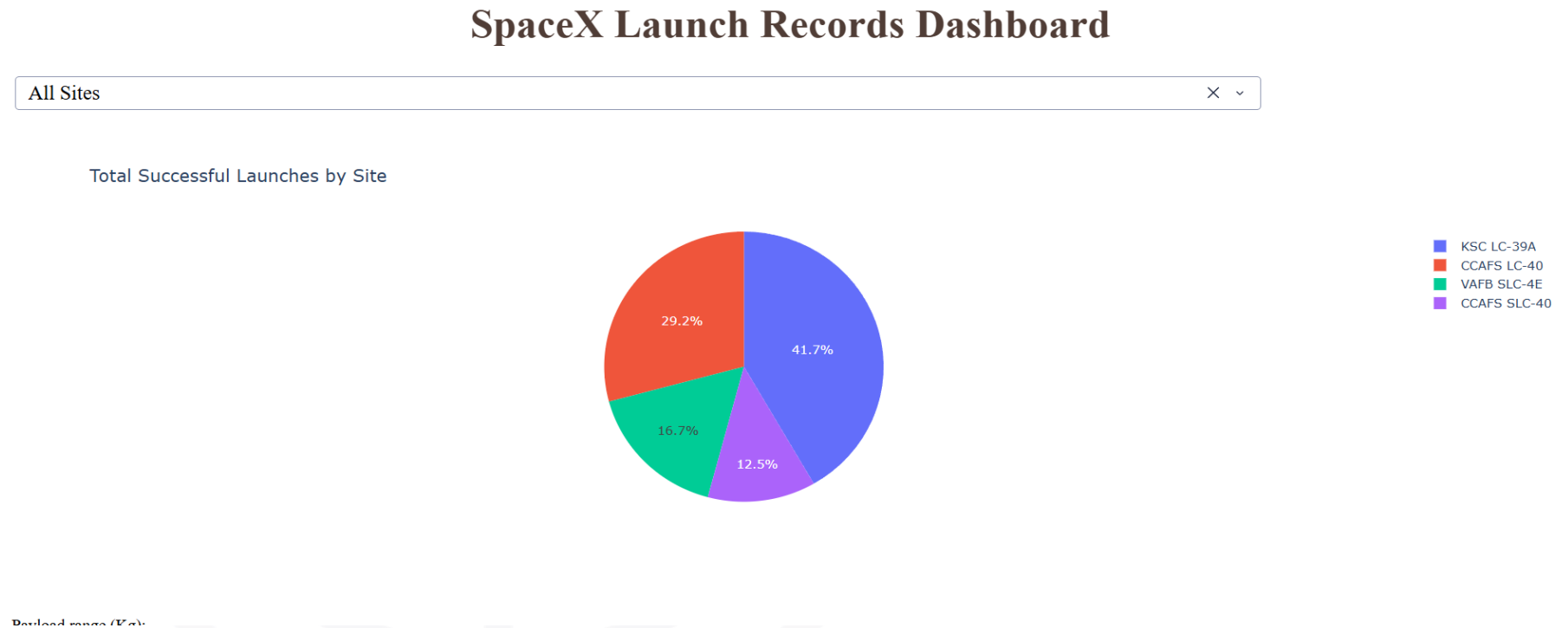
Map of SpaceX Launch Sites with Line to the Coastline



Map of SpaceX Launch Site with Line to the Closest City (Orlando)

DASHBOARD: Pie Chart of Successful Launches at Each Launch Site





https://github.com/rkdylan-manalo/IBM-Data-Science-Capstone-Project/blob/main/spacex-dash-app.py

Pie Chart from Dashboard showing Pie Chart of Successful Launches by Launch Site

Includes a Dropdown Menu to Highlight Specific Launch Sites and Interactive Pie Chart that highlights specific statistics

DASHBOARD: Scatter Plot of Payload & Booster Version

Takeaway: Allows Stakeholders to filter data by Payload Mass (kg) and Orbit Type in real-time to explore different launch scenarios.



Scatterplot from Dashboard Showing Payload Mass (kg) vs Booster Version

https://github.com/rkdylan-manalo/IBM-Data-Science-Capstone-Project/blob/main/spacex-dash-app.py

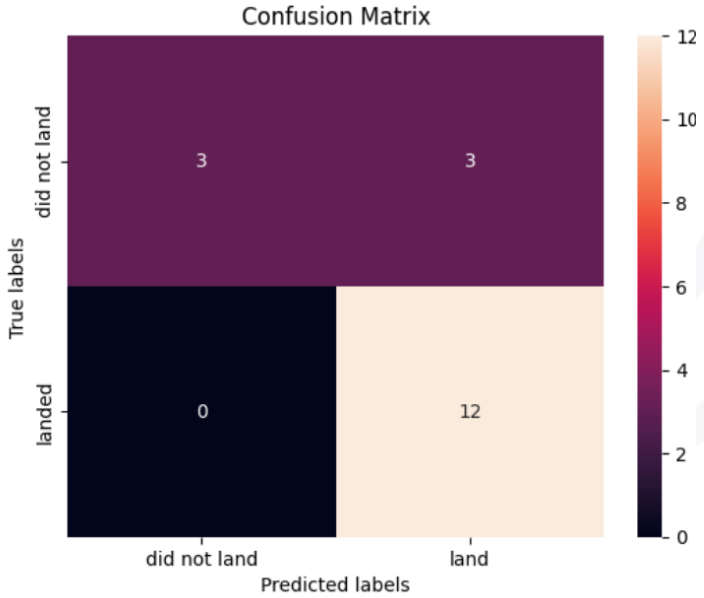
Includes interactive range slider and interactive scatter plot points

PREDICTIVE ANALYSIS: Machine Learning Model Results

Model	Training Accuracy	Test Accuracy
Logistic Regression	84.64%	83.33%
Support Vector Machine	84.82%	83.33%
Decision Tree	86.07%	83.33%
K-Nearest Neighbours	84.82%	83.33%

Table showing Machine Learning Model Training & Testing Accuracy

All Machine Learning Models had the Same Confusion Matrices (12 Successes, 3 Failures, 3 False Positives, 0 False Negatives



Takeaway: All four models were competitive in Test Accuracy, but Decision Tree Model comes out on top and achieved 83.33% Test Accuracy (86.07% Cross-Validation).

https://github.com/rkdylan-manalo/IBM-Data-Science-Capstone-Project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

DISCUSSION



- Implications: Reliable landing predictions to improve cost estimation & bid pricing.
- Launch Site Selection: Proximity to Coast/Equator & historical success should guide locations of future sites.
- Key Predictors:
  - Launch Number
  - Launch Site
  - Payload Mass
- Limitations:
  - Small sample size
  - Missing Payload Mass
  - Development of Technology over Time
- Business Value: Better predictors reduce financial risk for SpaceX and clients.

CONCLUSION



- Conclusion: We predicted the Falcon 9 first-stage landing success via API & Web Scraping and ML.
- Outcome: Decision Tree Model achieved 86.07% Training Accuracy; Launch Site & Number are key factors.
- Future Work:
  - Expanded dataset
  - Add Weather
  - Deploy Real World Dashboard
- Final Message: Data-driven insights can optimise launch decisions and reduce costs.

APPENDIX



- SpaceX APIs Data Sources:
  - getBoosterVersion: Rockets; URL: https//:api.spacexdata.com/v4/launches/rockets/
  - getLaunchSite: Launchpads; URL: https//:api.spacexdata.com/v4/launches/launches/
  - getPayloadData: Payloads; URL: url: https//:api.spacexdata.com/v4/launches/payloads/
  - getCoreData: Core; URL: https//:api.spacexdata.com/v4/launches/cores/
- Web Scraping Sources:
  - https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922
- Tools: Python, Pandas, Folium, Plotly Dash, Scikit-Learn